

Partial Differential Equations PhD Exam, January 2014

1. Let Ω be an open subset of $\mathbb{R}^n$ and let $u \in C^2(\Omega)$ be a harmonic function in Ω . Prove that u is analytic in Ω .

2. Let U be an open and bounded set in $\mathbb{R}^n$ with smooth boundary. Suppose u_1 and u_2 are both smooth solutions to

$$\begin{cases} u_t - \Delta u = 0, & \text{in } U_T := U \times (0, T], \\ u = g, & \text{on } \partial U \times [0, T]. \end{cases}$$

If $u_1(x, T) = u_2(x, T)$ for all $x \in U$, prove that $u_1 \equiv u_2$ within U_T .

3. Suppose u is a smooth solution to

$$u_{tt} - \Delta u = 0, \quad \text{in } \mathbb{R}^n \times (0, \infty)$$

and

$$u \equiv u_t \equiv 0 \quad \text{on } B(x_0, t_0) \times \{t = 0\}.$$

Let

$$C = \{(x, t) \mid 0 \leq t \leq t_0, |x - x_0| \leq t_0 - t\}.$$

Prove that $u \equiv 0$ within C .

4. Use characteristics to solve the following equation:

$$\begin{cases} x_1 u_{x_1} + 2x_2 u_{x_2} + u_{x_3} = 3u, & \mathbb{R}_+^3, \\ u(x_1, x_2, 0) = g(x_1, x_2), & \partial \mathbb{R}_+^3, \end{cases}$$

where g is a given smooth function.

5. Fix $\alpha > 0$ and let $U = B(0, 1)$ (open ball). Show there exists a constant C , depending only on n and α , such that

$$\int_U u^2 dx \leq C \int_U |Du|^2 dx,$$

provided

$$|\{x \in U \mid u(x) = 0\}| \geq \alpha, \quad u \in H^1(U).$$

6. Use Hopf–Lax formula to find a solution of

$$\begin{cases} u_t + \frac{1}{2}|Du|^2 = 0, & \mathbb{R}^n \times (0, \infty), \\ u = -\frac{1}{2}|x|^2, & \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

Is this solution unique in the weak sense?

7. Let U be a bounded subset of $\mathbb{R}^n$, $u \in W^{2,p}(U) \cap W_0^{1,p}(U)$, and $2 \leq p < \infty$. Prove that

$$\int_U |Du|^p dx \leq C \left(\int_U |u|^p dx \right)^{\frac{1}{2}} \left(\int_U |D^2 u|^p dx \right)^{\frac{1}{2}}$$

for some C independent of u .